

Hedi Bouassida

412-292-4857 | hedi.bouassida.pro@gmail.com | linkedin.com/in/hedi-bouassida | github.com/peroute

EDUCATION

Texas Tech University

Bachelor of Science in Computer Science – GPA: 4.0

Lubbock, TX

Aug. 2025 – Dec. 2027

- Relevant coursework: Object-Oriented Programming, Database Systems, Computer Organization & Assembly Language, Digital Logic, Discrete Mathematics

EXPERIENCE

Software Engineer Intern (AI/ML)

Jun. 2026 – Aug. 2026

AdReady AI

Remote

- Develop a multi-agent video ad QA platform that reviews AI-generated video ads for brands, flagging false claims, on-screen text errors, and storyline issues before ads ship
- Lead a 4-engineer product platform team charged with making the product usable end to end, owning user flows, review creation, uploads, product inputs, status/progress tracking, the results experience, application APIs, and authentication
- Contribute to system architecture for the multimodal extraction pipeline (speech transcription, OCR, keyframe analysis) and to the evaluation rubric scoring claims, CTAs, and on-screen text against product source documents
- Review teammates' pull requests as a repository code reviewer, enforcing codebase conventions and coordinating overlapping work across contributors
- Build frontend features on a Next.js, Tailwind, and Supabase stack; containerized the local development environment with Docker and the Supabase CLI

Programming Lead

Nov. 2025 – aug. 2026

Google Developer Groups on Campus, Texas Tech University

Lubbock, TX

- Led a Docker containerization workshop for 15+ members, creating presentations and hands-on demos to teach container workflows and best practices
- Coordinate technical workshops and coding sessions on API integration and emerging technologies

Assessment and Institutional Effectiveness Assistant

Apr. 2024 – May 2025

Community College of Allegheny County

Pittsburgh, PA

- Created analytical reports in Excel analyzing program outcomes for 10,000+ students across 5 academic departments; supported accreditation compliance evaluations by processing student performance data and identifying trends

Developer Intern

Apr. 2023 – Jun. 2023

Ezzayra

Tunis, Tunisia

- Developed a Follow-Line Robot on Raspberry Pi in Python, improving line detection precision by 33% via custom image-processing masks that enabled reliable autonomous navigation under varying lighting conditions

PROJECTS

Claim-ready (Hook 'Em Hacks 2026, solo) | Python, FastAPI, Gemini, CLIP, Whisper, ChromaDB

Apr. 2026

- Built a multimodal RAG pipeline solo in 24 hours that ingests a video walkthrough and a long PDF, then generates a structured, citation-backed output document (demo video)
- Engineered three retrieval layers: semantic text search over the PDF corpus using MiniLM embeddings, CLIP-embedded keyframe retrieval, and per-window Gemini Vision calls correlating narration transcripts with visual evidence
- Designed per-document ChromaDB collections with an evidence-linked schema where every detected item traces back to both its source timestamp and cited PDF clause; rendered final reports via Jinja2 and headless Chromium (Playwright)

Environmental Monitoring Platform | Python, FastAPI, PostgreSQL, Docker, React

Dec. 2025 – Present

- Developed station health monitoring system tracking online/offline status of 15+ weather stations, displaying real-time availability to farmers; integrated NASA POWER API enabling custom monitoring points on an interactive map for locations without physical stations
- Designed PostgreSQL schema for station metadata, health status, and historical weather data with efficient querying; containerized the full stack with Docker Compose

TTU Flashcard Generator | React, Flask, Python, Google Gemini API

Sept. 2025

- Built a full-stack application enabling students to generate course-based flashcards, achieving 10x faster study material preparation by injecting course objectives and learning outcomes into dynamic Gemini API prompts

CookBot | Python, YOLOv12, TensorFlow Lite, Android

Mar. 2025 – May 2025

- Trained and deployed a YOLOv12 object detection model with TensorFlow Lite for real-time ingredient recognition in an Android app, achieving 70% accuracy on 20+ classes over a 3,500+ image augmented training set

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript/TypeScript, HTML/CSS, Bash/Shell, Assembly

AI & ML: RAG, ChromaDB, CLIP, Whisper, Gemini/OpenAI APIs, PyTorch, TensorFlow/Keras, TFLite, YOLO

Backend & Frontend: FastAPI, Flask, PostgreSQL, Supabase, REST APIs, Jinja2, Pandas, NumPy, React, Next.js, Tailwind

Tools & Cloud: Git, Docker, Playwright, Jupyter, Linux (WSL, Ubuntu), Azure